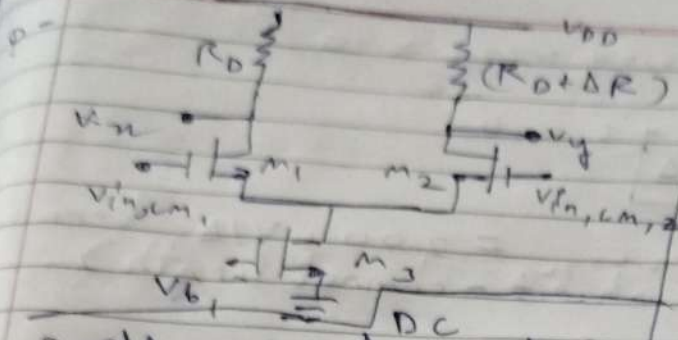




values used :- $\frac{1}{2} \left(\frac{W}{L} \right) = \frac{102}{0.15} \times \frac{1}{2}$

$V_{DD} = 1.8 \text{ V}$

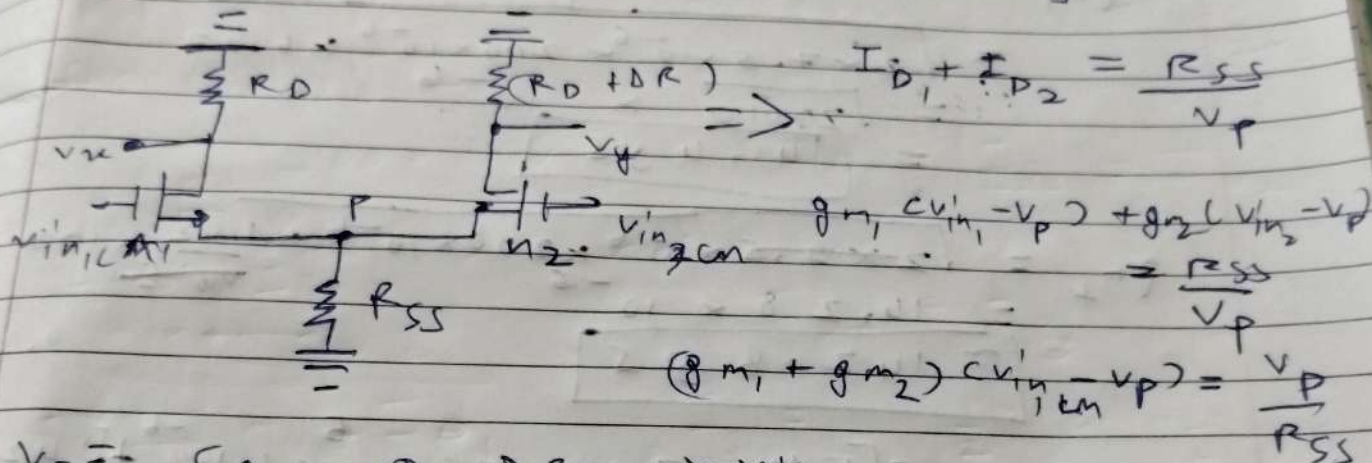
$V_{thn} = 0.65 \text{ V}$

$\mu_n C_{ox} = 420 \times 10^{-6}$

$g_{m1} =$

$g_{m2} =$

Small signal Analysis :- ($\beta = 0$) ($y = 0$)



$$v_P = \frac{(g_{m1} + g_{m2}) R_{SS}}{(g_{m1} + g_{m2}) R_{SS} + 1} \times v_{in,cm}$$

$$v_n = -I_{D1} R_D = -g_{m1}(v_{in,cm} - v_P) R_D$$

$$v_y = -I_{D2} (R_D + \Delta R) = -g_{m2}(v_{in,cm} - v_P) (R_D + \Delta R)$$

$$v_n - v_y = \frac{-g_{m1} \times R_D \times v_{in,cm}}{(g_{m1} + g_{m2}) R_{SS} + 1} - \frac{g_{m2} \times \Delta R \times v_{in,cm}}{(g_{m1} + g_{m2}) R_{SS} + 1}$$

$$I_{SS} = \frac{1}{2} \times \mu_n C_{ox} \times \frac{W}{L} \times (V_{GS} - V_{th})^2$$

$$= \frac{1}{2} \times 200 \times 6.67 \times (0.8 - 0.65)^2 \times 10^{-6}$$

Page No. _____
Date _____

$$I_{SS} = \frac{1}{2} \times 100 \times 6.67 \times 22.5 \times 10^{-10}$$

$$\frac{I_{SS}}{I_D} = \frac{1500}{15 \mu A}$$

we took $\mu_{n, \text{con}} = 200 \text{ pV/A}$ because of short-channel length.

$$\frac{W}{L} = 0.15$$

$$g_{m1} = \sqrt{2 I_D \mu_{n, \text{con}} \frac{W}{L}}$$

$$= \sqrt{2 \times 7.5 \times 10^{-6} \times 200 \times 10^{-6} \times 6.67}$$

$$= \sqrt{20.010 \times 10^{-6}}$$

$$= 14.14 \text{ pS}$$

$$g_{m2} = \sqrt{2 \times 7.5 \times 10^{-6} \times 200 \times 10^{-6} \times \frac{1.02}{0.15} \times 10^0}$$

$$= 14.28 \times 10$$

$$\approx 142.8 \text{ pS}$$

$$V_m - V_y = - \left[\frac{[14.4] \times 3.175 \times 10^5 \times 10 \times 10^{-6} + [142.8 \times 10^5 \times 10 \times 10^{-6}]}{e [184.2 \times 10^{-6}] \times 9.2 \times 10^5 + 1} \right]$$

$$= - \left[\frac{4.8895 \times 10^{-9} + 5.02 \times 10^{-9}}{170} \right]$$

$$V_m - V_y = \frac{0.9}{0.00170} \approx -0.005$$